Customer Churn Analysis: From Diagnosis to Recoverable Revenue

A decision driven investigation into churn drivers, customer risk, and retention strategy

1. Background & Overview

Customer churn is a significant commercial challenge for subscription-based businesses. Understanding where churn concentrates, which segments carry the highest risk, and what actions genuinely move the needle is critical to protecting revenue.

This project analyses subscriber, churn, and support-ticket data across ~500,000 telecom subscribers to answer five questions: Where is churn concentrated? Which segments sit meaningfully above baseline? What subscriber, pricing, contract, and support characteristics associate with churn? Who should retention teams prioritize? Where is the greatest financial opportunity?

The project combines Power BI, Power Query, SQL, and DAX each tool used for the question it answers best, not for its own sake. Power BI drove exploratory analysis, surfacing patterns across dimensions. SQL then answered the sharpest of those questions directly, calculating churn rate *within* a segment rather than a segment's share of total churn the version a reader can act on without doing mental math. Every major SQL finding corroborated the Power BI exploration; two findings that satisfaction rating predicts churn more consistently than issue type, and that tenure outperforms contract length as a retention lever only became clear because the analysis was run a second way.

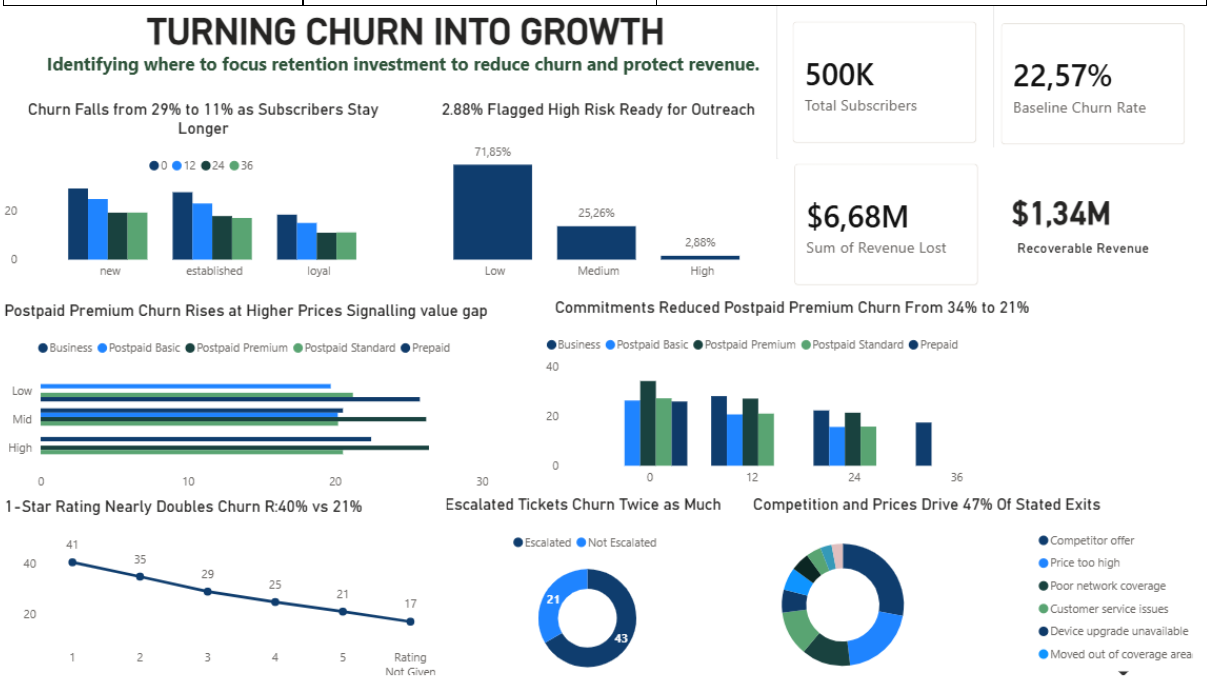
2. Decision

Should the business prioritize investment in early lifecycle onboarding, support experience quality, and price value realignment to reduce churn and which subscriber segments should retention teams target first?

No single recommendation is expected to resolve churn alone; each targets a distinct, independently confirmed driver. Their combined effect not any one lever is what's expected to meaningfully move the 22.57% baseline.

3. North Star Metrics

Metric	Value	What it tells you
Baseline Churn Rate	22.57% (112,843 of 500,261 subscribers)	The reference point every segment below is measured against
Active Subscribers at High Risk	2.88%	Who needs outreach right now, via the Risk Score
Estimated Monthly Revenue Lost	$6.68M	The recurring revenue no longer collected from churned subscribers, going forward
Estimated Recoverable Revenue	~$1.34M/month	If the top recommendations below close their targeted gaps



4. Recommendations

	Recommendation	Owner Team	Target	Business Reasoning
1	Structured onboarding for all new subscribers in their first 12 months, with a check-in at established tenure	Customer Success / Onboarding	Reduce No-Contract + New (29%) toward Established (27%), with Loyal (11%) as the long-term ceiling	Tenure, not contract length is what ultimately protects against churn; the earliest months are the highest leverage window to act
2	Shift support investment toward resolution experience broadly satisfaction and escalation handling, not billing tickets alone	Billing & Support Operations	Reduce 1-star (40%) and 2-star (35%) churn toward the 3-star rate (29%); reduce escalation rate	How an interaction *feels* predicts churn more reliably than what it was about fixing the experience fixes more than fixing any one ticket category
3	Review pricing and value proposition for Postpaid Premium (all charge tiers) and Prepaid Low-Charge subscribers	Product & Pricing Strategy	Bring Postpaid Premium (26.4%) and Prepaid-Low (25.8%) toward baseline	Risk sits at both ends of the price spectrum customers at the top feel overcharged, customers at the bottom feel underserved; this is a perceived-value problem, not an absolute-price one
4	Deploy the Churn Risk Score to drive proactive outreach to active High-Risk subscribers	CVM / Retention Marketing	Contact 100% of High-Risk active subscribers within 30 days of flagging	Converts a historical diagnostic into a live, operational tool the business can act on risk before it becomes a loss
5	Introduce contract commitment incentives across all	Sales / Loyalty	Reduce no-commitment churn (34% for Postpaid Premium)	Commitment meaningfully reduces churn, but never fully closes it alone it needs to complement, not

	plan tiers, prioritizing Postpaid Premium		toward the 24-month rate per tier	replace, Recommendations 1–3

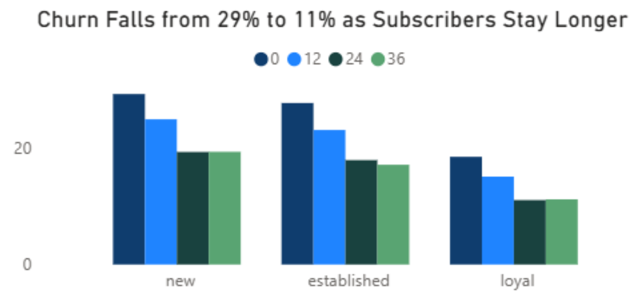
Acquisition channel was evaluated and deliberately excluded from formal recommendation, see Ruled Out.

5. Evidence & Insights Deep Dive

Each insight follows: **Finding → Business Meaning → What To Do About It.**

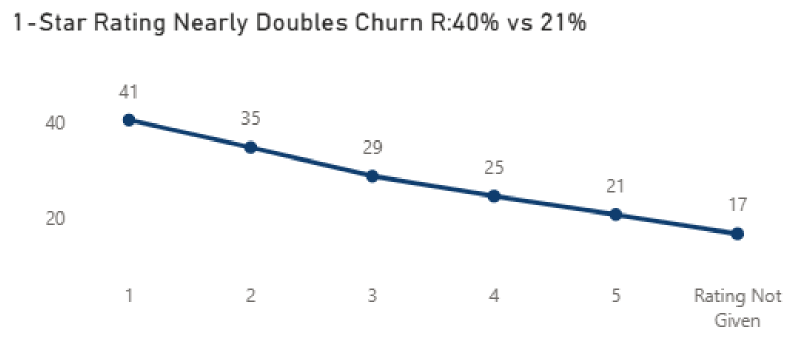
Onboarding: Recommendation 1

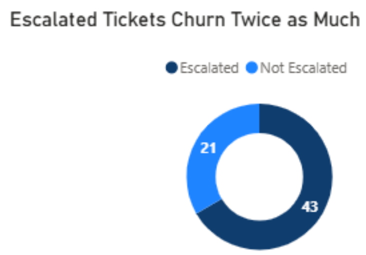
Finding: No-Contract + New subscribers churn at 29%, dropping to 27% once Established, and to 11% once Loyal (24+ months) a nearly 3x decline across the lifecycle, independent of contract status.
What it means for the business: Subscribers face two real windows of vulnerability an early trial period before the product proves itself, and a longer plateau where those who never committed to a contract never fully commit to the relationship. Tenure and habit, not paperwork, are what keep people.
What to do: Concentrate onboarding investment in the first 12 months, with a lighter secondary check-in at the established-tenure mark, rather than spreading effort evenly across the full subscriber lifecycle.



Support Experience: Recommendation 2

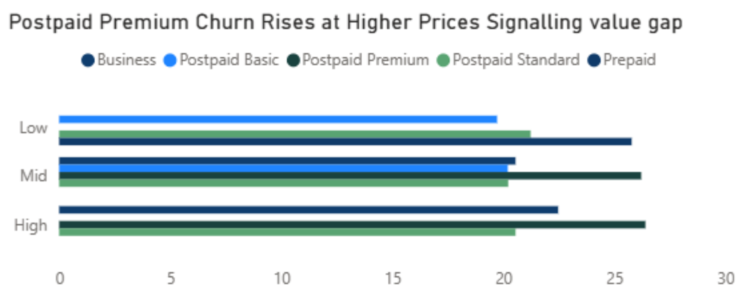
Finding: Churn tracks satisfaction in a smooth gradient 40% at 1-star down to 21% at 5-star regardless of whether the ticket was about billing, network, or devices. Escalated tickets independently show a 43% vs 21% churn gap.
What it means for the business: The subject of a support ticket matters less than how the interaction felt. Fixing resolution quality broadly will do more for retention than routing extra attention to any single-issue category, including billing.
What to do: Redirect support quality investment toward resolution experience and escalation handling as the primary lever, with billing-specific fixes as a secondary, not primary, focus.

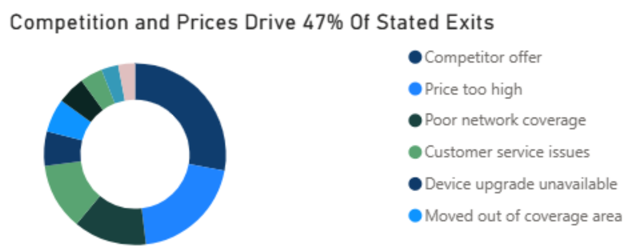




Pricing: Recommendation 3

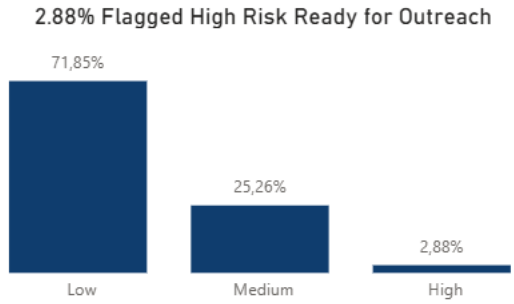
Finding: Postpaid Premium churns at 26.4% (High-charge) and 26.2% (Mid-charge); Prepaid Low-charge churns at 25.8%. Stated churn reasons independently confirm this: competitor offers (27%) and price too high (20%) account for the largest share of stated exits, with poor network coverage (13%) a distant third.
What it means for the business: Risk isn't about being expensive or cheap in absolute terms it's about whether the price matches what the subscriber feels they're getting. Both extremes lose people, for opposite reasons.
What to do: A pricing/packaging review specifically for these two segments, not a blanket price change the fix is about perceived value, not the number on the invoice.





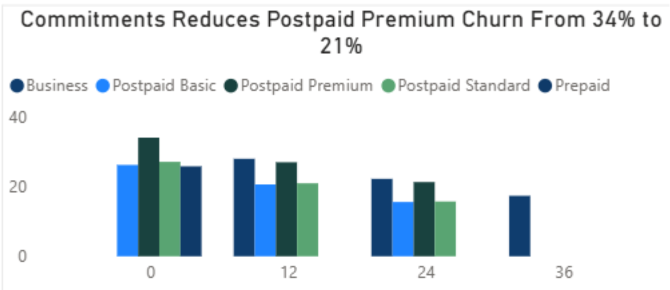
Risk Score & Deployment: Recommendation 4

Finding: A weighted 5-flag score (No Contract + New Tenure, Billing Dispute, Escalated Ticket, Low Satisfaction, Price-Value Mismatch each independently confirmed, unlike the excluded channel flag) identifies the 2.88% of active subscribers carrying the highest combined risk. High-Risk subscribers churn at a substantially higher real rate than Low-Risk, confirming the score is genuinely predictive.
What it means for the business: The score is precise (correct when it flags someone) but narrow in coverage it identifies a high confidence subset of churners, not the majority of churn volume. It's a targeted efficiency tool, not a standalone fix.
What to do: Deploy for proactive, individualized outreach alongside not instead of the structural recommendations above.



Contract Commitment: Recommendation 5

Finding: Contract commitment reduces churn consistently across every plan tier. Postpaid Premium starts highest with no commitment (34%) and falls furthest by 24 months (21%); Business follows a similar curve but sits lower throughout (28% → 22% → 17%).
What it means for the business: Commitment works, but no tier reaches baseline through commitment alone even the most contract responsive segment still churns above the company average at 24 months.
What to do: Use commitment incentives as a complement to onboarding and pricing fixes, not a substitute measure success against each tier's own trajectory, not against baseline directly.



6. Estimated Financial Impact

To size the commercial significance of each recommendation, targeted SQL analysis defined the addressable subscriber population, calculated its average monthly charge, and compared its observed churn rate against a realistic target rate generally the next best performing segment within the same population, not an arbitrary goal.

Estimated monthly revenue at stake = Addressable population × Churn-rate gap × Average monthly charge

Recommendation	Segment / Target	Est. Monthly Revenue at Stake
1. Onboarding	No-Contract + New → Established rate	~$57,100 near-term, up to ~$348,500 under the full Loyal-tenure target
2. Support Experience	1-star & 2-star tickets → 3-star rate	~$749,300
3. Pricing	Postpaid Premium (High) + Prepaid (Low) → baseline	~$402,200
5.Contract Commitment	Postpaid Premium, no-commitment → 12-month rate	~$134,750 (~$1.62M annualized)

These figures represent potential monthly recurring revenue at stake, not guaranteed recovery. Segments may overlap a subscriber can appear in more than one recommendation's target population so these should be read as independent opportunity sizes, not summed into a single guaranteed total. The underlying analysis is observational; the estimates are directional, not causal forecasts.

7. Data quality

Data quality checks performed:

- All three source CSVs imported as text; every numeric comparison explicitly cast before calculation
- Churned and Escalated binary flags verified against known totals before being trusted (not assumed) an early pass had Churned reversed
- Two segment combinations (Plan Type × Charge Group, n=2 each) excluded from reporting after producing a misleading 50%+ churn rate on a statistically meaningless sample
- Blank Satisfaction Ratings retained as their own labelled category rather than silently dropped, to reflect the true population

SQL views (8 total, static and independent not part of the relational model, each answering one targeted question directly): Contract × Tenure, Satisfaction Rating, Issue Type × Satisfaction, Plan Type × Charge Group, Churn Reasons, Acquisition Channel × Tenure, Contract Commitment, Escalated Tickets.

8. Caveats & Assumptions

- **Static dataset** — a historical snapshot, not a continuously refreshed production feed
- **Observational, not causal** — relationships identified (satisfaction, escalation, pricing) are strong associations, not proof that changing them *causes* churn reduction
- **Risk Score is rule-based, not statistically calibrated** — flag weights reflect observed churn gaps, not a fitted model; a logistic regression is a natural next iteration
- **Segment overlap** — a subscriber can appear in more than one recommendation's target population, so financial estimates should not be summed into one guaranteed total
- **"Significant" segments** were defined using a fixed 5-percentage-point / 1.5x-baseline threshold, not a formal significance test
- **Financial estimates are directional** — they depend on the selected target/comparison churn rate and should be treated as potential exposure, not a committed forecast

Ruled Out: No Independent Signal

Explicitly tested and excluded: device protection (~22% churn, at baseline), support contact channel, usage behaviour, payment punctuality, and acquisition channel (churn was consistent across channels once tenure was controlled for a real but secondary pattern, not worth a standalone recommendation or a Risk Score flag).

Dataset note: this analysis uses a synthetic telecom dataset (~500,000 subscribers), not real customer or company data. The findings demonstrate analytical methodology, tool proficiency, and decision driven diagnostic thinking, rather than real market insight the specific churn drivers, financial estimates, and recommendations should be understood as illustrative of the analytical process, not claims about any actual company's subscriber base.